

active community of users, many of whom began contributing to its vast and growing pool of user-created resources.

As a result, Linux developers can access countless libraries of pre-written utilities and functional blocks, created by fellow users. With many of the low-level elements of their applications already available, developers are able to focus their efforts on the proprietary functions that add value to their product and differentiate it from its competitors. Linux's ability to shorten a product's time to market while reducing development costs made it a top choice for the embedded systems that run 'under the hood' of thousands of consumer and industrial products. In addition, Linux powers the servers that run 96.5 per cent of the top one million domains in the world (as ranked by Alexa), 92 per cent of Amazon's 350,000 EC2 Web servers, and all 500 of the world's most powerful supercomputers (<https://www.makeuseof.com/tag/linux-market-share/>).

Phase 2: Arduino

The Arduino embedded computing platform was at the vanguard of the second open source wave (hardware) when it was first conceived in 2003 as a student project at the Interaction Design Institute Ivrea in Ivrea, Italy. The project's audacious goal was to develop a very low cost, easy to program, single-board computer that would enable designers, artists and other non-engineers to "...build digital devices and interactive objects that can sense and control objects in the physical and digital world" (<https://www.arduino.cc/en/Guide/Introduction>). In retrospect, the developers significantly underestimated the project's actual impact.

The original Arduino board was roughly the size of a business card and provided easy access to the signal lines of its 8-bit microcontroller through a series of pins along each side of the board. These pins allow a developer to easily connect the processor's input and output lines to a real-world device, such as a

light, sensor, or motor. They can also be used to connect drop-in daughter boards (known as shields) equipped with sensors, network interfaces, or other functionality. Programming the boards is accomplished using simple, English-like commands that are written and compiled within a user friendly integrated development environment (IDE).

Arduino's low cost, versatility, and ease of use led to its explosive popularity with the technology hobbyists who embraced the do-it-yourself (DIY) philosophy of the so-called Maker movement. Sold in the millions, the tiny open source board and its variants served as the silicon intelligence that enabled hobbyists to unleash their imaginations and create things like low-cost automated telescopes, smart clothing, personal robots, and automated home brewing systems.

Meanwhile, as enterprising makers began to offer innovative commercial products on Kickstarter and other crowdfunding sites, Arduino technology helped them bring their ideas to market quickly and inexpensively.

A powerful legacy: As some of the earliest open source platforms, Linux and Arduino were instrumental in upending the conventional top-down business model in which proprietary technologies were developed in isolation by large corporations and doled out to those who could afford them. In contrast, opensource's bottom-up approach enables designers to harness the collective intelligence of a user community to create tools and products that reflect real-world needs.

The overwhelming success of Linux and Arduino also demonstrated that product differentiation on a common, non-proprietary platform is not only possible, but desirable.

Open source technologies perform another vital and surprising role within the tech economy by serving as a catalyst that enables the creation of other open source technologies that make new and unexpected applications possible. In much the same way that Stallman's GNU

project provided many of the concepts and enabling technologies that made Linux possible, Arduino gave makers, tinkerers, and would-be entrepreneurs the low-cost, easily hackable controller they'd need to tackle projects that might have seemed unthinkable until then.

Among other things, the humble controller board provided the spark that ignited the 3D printing revolution. Although the first 3D printers were developed in the 1980s, they were still costly and relatively rare, typically found only in large corporate R&D labs. This changed in 2005 when a team at the University of Bath in UK began a project to develop a low-cost 3D printer that could print most of its own components. They named the project RepRap, short for *replicating rapid prototyper*. The machine would melt inexpensive plastic filament and extrude it through a print head that would be moved across the X, Y and Z axes using inexpensive stepper motors. Each of these motors was controlled by its own Arduino board, while other Arduinos took care of various housekeeping functions (<https://www.digikey.com/en/articles/techzone/2011/nov/arduino-open-source-platform-unleashes-creativity>).

An early RepRap prototype printed its first useable part around a year later, paving the way for a subsequent design that was able to print more than half of its functional parts in February 2008. Tech enthusiasts from around the world quickly embraced the RepRap project, using the open source design to build their own machines, and contribute improvements that they'd made. Soon, entrepreneurs were using RepRap's 'genetic code' as the basis for affordable commercial products, such as the Thing-o-Matic, a sub-US\$ 1000 3D printer kit, produced by MakerBot. The Thing-o-Matic, and virtually all of the other early machines, used Arduino controllers, and most of the low cost units sold today use Arduino-style controllers.

The Arduino also played a seminal role in the development of low-cost multi-rotor and fixed-wing drones that enjoy wild popularity with both hobbyists